

available with Swatch.

Option	A brief description
<code>--config-file filename</code>	Runs Swatch using the specified file name as a configuration file. If no file name is given, use the default one.
<code>--restart-time time</code>	Restarts Swatch at the indicated time.
<code>--daemon</code>	Runs Swatch as a system daemon.
<code>--examine file</code>	Makes Swatch do a complete pass through the specified file.
<code>-tail file</code>	Swatch reads only the newly added lines in the file.
<code>--help</code>	Displays a short help summary.
<code>--version</code>	Displays the version of the Swatch script.

For example, the following command...

```
./swatch --config-file /home/wayam/my-swatch-config -- daemon
```

...will run Swatch using the configuration file found at `/home/wayam/my-swatch-config` instead of the default `config` file. It will also run it as a background process or daemon. Please note that the above options may be issued alone or together.

Swatch expects the `.swatchrc` file to reside in the home directory of the user who invokes Swatch. Swatch also keeps its temporary files there by default. Each time it's invoked, it creates and runs a script called a 'watcher process', the name of which ends with a dot, followed by the PID of the Swatch process that created it. Swatch generally doesn't clean up after itself very well; it tends to leave watcher-process scripts behind. Thus, users are expected to keep an eye out and periodically delete these in their home directory.

The command `--tail-file=[path to log file]` directs Swatch to watch a specific log file for potential matches.

An interesting feature of Swatch is that one can run multiple instances of it, each configured to use a customised configuration file and watch a different log file. Some of the common log files that Swatch uses are listed below.

- `/var/log/maillog` - logs all e-mail messages
- `/var/log/cron` - logs messages regarding cron job schedules
- `/var/log/messages` - logs system messages

Let us look at all the actions Swatch can take when a specific search is completed. Swatch provides a variety of actions that you can perform in response to a matched event. One can output alerts to the console, pipe output to another log file, send email alerts, or even execute a remedial script.

As more than one action can be applied to a single event, you can combine these to suit your requirements. The following table illustrates various Swatch action statements.

Swatch action statement	A brief description
<code>echo mode</code>	The search text can be echoed onto the screen. Please note that <i>mode</i> is optional and it indicates the colour in which it is to be displayed.
<code>bell number</code>	This rings the PC internal speaker the number of times indicated by <i>number</i> .
<code>exec command</code>	This executes a command line parameter. One can configure this to call a script that can take further action.
<code>pipe command</code>	This passes along a command to another process.
<code>write user1: user2</code>	This causes an alert to be sent via a UNIX write command, and it can be sent to one user or a group of users.
<code>mail addresses=address1:address2:address3,subject=text</code>	Sends an email using the Sendmail program to a single email address or multiple email addresses, separated by colons.
<code>throttle HH:MM:SS</code>	Waits for HH:MM:SS (a period of time) after a line triggers a match, before performing actions on another match of the same expression.

As can be seen from the above table, Swatch can notify you of flagged log events in several different ways. The easiest is to just have it beep or echo on the screen. If you are not around the server all the time, then you can have it email you. If your pager or cell phone supports text messaging via email, you could have it send the message directly to you. You can also write a script to have the server dial a pager number using the UNIX *tip* command.

Echoing the log output is considered as one of the most basic functions of Swatch. This can be utilised as a way to gain the attention of users (if they are currently using the console) by outputting the log contents to the console.